

# Object Detection Extension

<https://github.com/aiyshwary/Object-Detection-Extension-in-Rapid-Miner>

[Python](#)

[PyTorch](#)

[ONNX Runtime](#)

[RapidMiner](#)

[Object Detection](#)

## OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment, supporting fine-tuning and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

## IMPACT

Developed two RapidMiner operators for object detection — `FineTuneObjectDetectionModel` and `ObjectDetectionModelInference` — enabling users to train and deploy detection models within a visual workflow without writing code.

## KEY DETAILS

`FineTuneObjectDetectionModel` accepts image and annotation directories (YOLO/Pascal VOC)

- and outputs model weights and class list CSV.

`ObjectDetectionModelInference` runs predictions on new images and returns bounding boxes and

- class labels as a `DataFrame`.

Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying

- speed-accuracy trade-offs.

Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in

- RapidMiner Studio.

Enables domain-specific object detection with minimal ML expertise using a drag-and-drop

- process designer.